Build a Paper Prototype

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Build a Paper Prototype

> Prepare Build a Paper Prototype: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Build a runnable paper **prototype** with explicit **instruction**s and visible state

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Design the Decision Process

Without opening the old lesson, explain one key idea from Design the Decision **Process**. Give an example and a mistake that the idea helps you avoid.

Look for: Design a decision **process** that covers valid, invalid and no-match requests

Reference example, not the only acceptable wording: Clarification follows invalid or missing **input**. A suggestion follows the first candidate passing both compatibility conditions.

No match follows a valid request after all candidates fail. A single failed candidate does not settle whether later candidates fit, so the **process** must advance.

Find a Real Problem

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

Look for: Describe a specific user difficulty using supplied **evidence** without inventing its cause

Reference example, not the only acceptable wording: For example: fictional pupils choosing activities in the supplied classroom exercise had to choose again in four of ten requests because the first choice did not fit the time or room conditions.

Build an AI app names a solution or construction task without specifying a user difficulty or **evidence** of need.

Prompts Are Design Instructions

Without opening the old lesson, explain one key idea from Prompts Are Design **Instruction**s. Give an example and a mistake that the idea helps you avoid.

Look for: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements

Reference example, not the only acceptable wording: Possible omissions include audience, source facts, required fields, format, missing-value behaviour and limits on extra claims.

C supplies **evidence**, Grade 6 specifies readers, and four labelled lines define visible fields and structure. These requirements support checks but do not guarantee compliance.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> Assign three people as **input** reader, rule follower and **output** recorder. Run the repaired sorter on paper, metal and an unreadable description. Record expected and observed outcomes. What does this **prototype** demonstrate?

Concept to explain

A **prototype** tests the idea before polish or technology.

Term	Meaning
prototype	early representation used to test an idea
low fidelity	simple version focused on function
simulation	acting out system behavior
operator	person executing the prototype process

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

This session builds a runnable paper implementation of the existing rule-based recommender. A **trace** reports execution; the specification and manual baseline determine the **expected result**.

Do not silently coach an **operator** through omitted **instruction**s. Distinguish an **instruction** defect from a failure to follow a correct **instruction** by comparing the saved cards with the **trace**.

A paper **prototype** can test logic and handover, but does not establish automated speed, real pupil satisfaction or general reliability.

Readiness answer

Advance to the next catalogue row. No match is justified only after valid **input** and failure of every row. A passing row produces the first suggestion and stops the run.

Paper for request, **instruction** and catalogue cards, using the exact B version 1 values in the lesson

A movable marker and **trace** sheets with version labels and check columns

The Week 30 requirements and manual checklist; a folder for unchanged runs and revisions

End goal for the pictorial case

By the end, I can build and operate a paper **prototype** while recording actual **trace**s.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Build a Paper Prototype using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT paper item card

2. Process — the deciding step

PROCESS partner executes the **process** without improvising

3. Output — the result

OUTPUT paper **output** card and **observation** notes

Key idea

CORE IDEA A **prototype** tests the idea before polish or technology.

How the mechanism works

UNDER THE HOOD A low-fidelity **prototype** externalizes assumptions cheaply.

Wizard-of-Oz **simulation** is valid when the human **operator** follows the specified **process** and does not secretly add judgment unavailable to the proposed system.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE One student presents an item card while another follows the written sorter **process** exactly and returns the documented **output**.

Step 1

STEP 1 — OBSERVE paper item card

Step 2

STEP 2 — **TRACE** partner executes the **process** without improvising

Step 3

STEP 3 — CONCLUDE paper **output** card and **observation** notes

Model answer

MODEL REASONING The **operator** must use only documented **input**s and steps. Any question, pause, or improvisation reveals a missing rule, unclear label, or unsupported **input**. Revise the paper flow before adding technology.

A claim to avoid

DO NOT OVERCLAIM A helpful **operator** can accidentally make a weak design look successful by using unstated human knowledge.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

A paper **prototype** has **input** cards, decision cards and paper, plastic and review trays.

How can a learner simulate it?

Exact explanation to model

Why the result follows

Read **input**, follow written decisions and place a token in a tray. Record the **trace**. Human **simulation** does not establish model **training**.

Explain the added worked example

Explain the added worked example · 1

This week you will turn the decision **process** into a paper **prototype**. One partner operates the **instruction**s and another records the steps.

A working **prototype** needs more than a polished drawing: someone else must be able to follow it without asking its author to supply missing rules. Keep the version you ran so the **trace** can be checked against it.

Explain the added worked example · 2

Copy this catalogue in order onto four cards: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No.

These are fictional paper-model values with materials assumed available. Label the catalogue B version 1.

Explain the added worked example · 3

Keep the Week 30 contract: **input** is a whole number of available minutes from 0 to 30, and Quiet required Yes or No.

An activity fits when its time is no more than the available minutes AND either quiet is not required or the activity is quiet. No means either kind is allowed. Suggest the first fit with a reason; otherwise report no match.

Missing or invalid **input** needs clarification. Suggestions do not book materials or publish anything.

Explain the added worked example · 4

Make a request card with Available minutes and Quiet required fields, a marker for the current catalogue row, and a **trace** sheet. Head the sheet Run ID, **prototype** version, catalogue version and request.

Below it use columns Step, current row, time check, quiet check and next action or **output**. Use not checked when validation stops the search.

Explain the added worked example · 5

Write **instruction** cards P1 version 1. Card 1: validate both fields; if invalid, name the field needing correction and stop; otherwise put the marker at B1.

Card 2: check the marked row against both compatibility conditions; record both checks. If both pass, suggest that row with the reason and stop.

Card 3: if either fails and another row remains, advance exactly one row and return to Card 2; if no row remains, report no match and stop. Begin each new request at Card 1 with a cleared **output**.

Key terms

Term	Meaning
Prototype	An early working version used to explore and test a design
Instruction	A precise action or decision for the operator to follow
Trace	A record of the steps and state during one run
Expected result	The result required by the agreed specification
Observed result	What actually happened when this version was run

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 6

Before operating P1, use the full manual checklist to write the **expected result** for 9 minutes, Quiet required No. Then run the cards literally. Do not replace a surprising result with the expected one.

Recording disagreement gives you something to investigate; hiding it removes useful **evidence**.

Trace step for 9 minutes No	Checks recorded	Next action or output
1 Validate	Whole minutes in range; No accepted	Place marker at B1
2 Check B1	10 is greater than 9; quiet unrestricted	Move to B2
3 Check B2	8 is no more than 9; quiet unrestricted	Suggest B2 with reason; stop

Model the reasoning aloud · 7

This **trace** agrees with the expected B2 result. It shows that this run followed the **instruction**s. One matching result does not prove that every request will work or that the **prototype** is faster than the manual checklist.

This is a system of **written rule**s, not a model trained on examples.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Build a Paper Prototype

A fictional team turns a storage-bin specification into movable **instruction** cards.

The facts we will use

> **Supplied facts**
>
> Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.

> **Supplied facts**
>
> For request 4, write baseline Y before operating cards. **Prototype** cards mistakenly stop with X immediately.

AI ACADEMY / AI-1/32

Build a Paper Prototype

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Expected result

Baseline checks X then Y and expects Y for count 4.

2. Operate literally

Follow the current prototype cards without secretly repairing them.

3. Actual trace

Prototype stops at X; it never performs the capacity comparison.

4. Record mismatch

Expected Y, actual X: locate the missing check.

CHANGE ONE THING

Insert "if capacity is below count, continue" before accepting a bin.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Expected result	Baseline checks X then Y and expects Y for count 4.
2. Operate literally	Follow the current prototype cards without secretly repairing them.
3. Actual trace	Prototype stops at X; it never performs the capacity comparison.
4. Record mismatch	Expected Y, actual X: locate the missing check.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

What this example does not establish

Keep the stated limit

One repaired paper case does not establish usability, correctness on every **input** or performance of a later software implementation.

Change one thing in the pictured case

Change one thing

Change: Insert “if capacity is below count, continue” before accepting a bin.

Predicted result

Prediction: The revised **trace** rejects X and accepts Y for 4.

Why the result follows

Reason: 3<4 is true; 7<4 is false, so Y fits.

Ask while showing the picture

Ask while showing the picture · 8

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Build a Paper Prototype, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A paper **prototype** has **input** cards, decision cards and paper, plastic and review trays.

2. State the decision to explain

Write the question to explain: How can a learner simulate it?

3. Reveal the worked reasoning

Reveal after **prediction**s: Read **input**, follow written decisions and place a token in a tray. Record the **trace**. Human **simulation** does not establish model **training**.

Hold back the independent task

Keep for an independent attempt: Your paper **prototype** shows an **input** card, a decision card and an **output** card. Explain what a person must simulate and what the **prototype** cannot prove.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, **trace** conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

8 Yes is valid. B1 fails time 10 greater than 8 but passes quiet; B2 passes time 8 equal to 8 but fails quiet; B3 fails time 12 greater than 8 but passes quiet; B4 fails time 15 greater than 8 and fails quiet.

After B4, no match. Blank quiet stops at Card 1 with clarification and zero candidate checks. Exit: starting at B2 violates P1's reset **instruction**, so this is an execution mistake.

Keep the original **trace** and run P1 again from validation and B1; for 11 No, suggest B1 because 10 fits 11 and quiet is unrestricted.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the conditions for continuing a search and reporting no match.

Stage	Teacher action
7 to 20 minutes	Read the contract, build and label the P1 kit, and complete W1.
20 to 33 minutes	Model the 9 No trace, keeping expected and observed records separate.
33 to 45 minutes	Partners complete W2 with an unaided operator and retain any disagreement.
Session 2 start to 8 minutes	Compare the stale-marker execution mistake with the P0 instruction defect.
8 to 23 minutes	Complete W3 and the guided 8 Yes trace; record correction versions without altering old runs.
23 to 38 minutes	Collect W4 independently, including cards and two trace records. Offer support only after the first attempt.
38 to 45 minutes	Complete W5 and the exit; save the handover kit for routine tests next week.

Reduce help deliberately

Completion practice

Completion task: Use the manual checklist to predict P1's **output** for 11 minutes No. Then ask a partner to operate your saved cards.

Record the **expected result** separately from the observed steps and **output**, including the reason. Explain why the **operator** should stop without checking later rows.

Reasoning to listen for

Teacher check: The expected **output** is B1: 10 is no more than 11, quiet is unrestricted and B1 is first. A correct observed **trace** validates, starts at B1, passes both checks and suggests B1 with this reason.

Stop because the specification selects the first fit. If a pupil observes a different run, retain and investigate that **evidence** rather than replacing it with this expected **trace**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full **trace** only when the partial **trace** is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Building and tracing a paper **prototype**. Proposed explanation: “A correct-looking answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **Expected repair**
>
> Expected correction: Inspect the request, versions and **trace** to check how it was reached.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> What should appear in the test log before and after the repair?

Reasoned teacher answer

Keep original version: expected Y, actual X, missing comparison. Add revised version: expected Y, actual Y, with its full **trace**.

Changed-case reasoning: The revised **trace** rejects X and accepts Y for 4. 3<4 is true; 7<4 is false, so Y fits.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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Run the weekly evidence lab

> Run the Build a Paper Prototype investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Run the sorter with a partner acting as the computer.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the **expected result** separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR A paper **prototype** is already a trained AI system.

For guessing, return to given facts and request a **trace**. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A correct-looking answer is enough: Inspect the request, versions and **trace** to check how it was reached.

The observer should repair missing **instruction**s during a run: Record the ambiguity, revise a labelled version and rerun separately.

An **observed result** should be changed to match the expected one: Preserve disagreement as **evidence** and investigate its cause.

A successful paper run proves a trained AI works: These are explicit rules; one run supports only a limited claim about that implementation.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Build a Paper Prototype questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Read **input**, follow written decisions and place a token in a tray. Record the **trace**. Human **simulation** does not establish model **training**.

Independent question

Independent question · withhold the key

Your paper **prototype** shows an **input** card, a decision card and an **output** card. Explain what a person must simulate and what the **prototype** cannot prove.

Expected independent answer

Teacher key · Expected independent answer

A person must follow the decision **process** and produce an **output** for each **input**. The paper model can test workflow clarity but cannot prove that a real trained model will be accurate.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions

Criterion	0 points	1 point	2 points
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A paper **prototype** is already a trained AI system.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Expected outcomes are P, M, review. Record the actual **trace** and any mismatch.

The activity demonstrates whether people can follow the specified **process**; it does not show that a model has learned, nor that a future automated implementation is accurate.

Explanations for the additional workbook examples

W1 Assemble the runnable kit

Workbook question · W1

Create P1 version 1 using the three **instruction** cards in the lesson. Copy catalogue B version 1 and make a request card, current-row marker and **trace** sheet. List what each item records.

Explain which information stays fixed during one run and what changes.

Teacher reasoning · W1

The kit contains labelled **instruction**s and catalogue, request fields, a marker and a **trace** sheet with run and version labels. The request and catalogue stay fixed during the run; the current row and **trace** advance.

The final **output** starts clear. The cards validate, check both conditions, return the first fit, advance after failure and stop with no match after exhaustion.

W2 Run and record the first match

Workbook question · W2

Use the manual checklist to predict P1's **output** for 11 minutes No. Then ask a partner to operate your saved cards. Record the **expected result** separately from the observed steps and **output**, including the reason.

Explain why the **operator** should stop without checking later rows.

Teacher reasoning · W2

The expected **output** is B1: 10 is no more than 11, quiet is unrestricted and B1 is first. A correct observed **trace** validates, starts at B1, passes both checks and suggests B1 with this reason.

Stop because the specification selects the first fit. If a pupil observes a different run, retain and investigate that **evidence** rather than replacing it with this expected **trace**.

W3 Locate the source of a wrong result

Workbook question · W3

Saved **instruction**s P0 say “start at B1; if B1 fails, report no match.” For the valid request 9 minutes No, an **operator** follows P0 exactly and reports no match. Identify the **instruction** defect and write the changed step.

Explain how to label the correction and what to retain from the failed run.

Teacher reasoning · W3

P0 stops after one failure and misses B2, which fits 9 No. After failure, advance to the next row and continue; no match follows only after all rows fail. Save a new version with the change recorded.

Keep the original P0 **instruction**s, request, expected B2, observed no match and **trace**. Rerun the corrected version and record a new result.

W4 Build a new route through a small catalogue

Workbook question · W4

Independently make and **trace instruction** cards for this paper equipment catalogue: E1 first, capacity 3 people; E2 second, capacity 6. Valid **input** is a whole group size from 1 to 8.

Suggest the first item whose capacity is at least the group size; no booking occurs. Include invalid **input** and no match. Write **expected result**s, then operate your cards for groups of 5 and 7.

Record both runs and explain every stopping point.

Teacher reasoning · W4

Validate whole group size 1 to 8, otherwise clarify and stop. Start E1; if capacity is at least the group size, suggest and stop; otherwise advance to E2. If E2 fails, no match and stop.

Expected 5: E1 fails 3 below 5; E2 passes 6 at least 5, suggest E2. Expected 7: both capacities too small, no match. Record actual runs separately and investigate any disagreement.

W5 Make the evidence repeatable

Workbook question · W5

A teammate keeps only a final “B2” answer and throws away the request and old **instruction**s after editing them. Explain why another person cannot check this result.

List what to save for each run and explain why an observed error must remain in the record.

Teacher reasoning · W5

B2 alone does not identify the request, version or route taken, so its correctness cannot be checked.

Keep run ID, exact request, **instruction** and catalogue versions, **expected result**, observed steps and **output**, and any change with its later rerun.

Keeping errors allows the team to locate and verify repairs instead of showing only favourable results.

Assess independent transfer

Independent question · withhold the key

Selected case: Independently make and **trace instruction** cards for this paper equipment catalogue: E1 first, capacity 3 people; E2 second, capacity 6. Valid **input** is a whole group size from 1 to 8.

Suggest the first item whose capacity is at least the group size; no booking occurs. Include invalid **input** and no match. Write **expected result**s, then operate your cards for groups of 5 and 7.

Record both runs and explain every stopping point.

Selected case answer

Teacher key · Selected case answer

Validate whole group size 1 to 8, otherwise clarify and stop. Start E1; if capacity is at least the group size, suggest and stop; otherwise advance to E2. If E2 fails, no match and stop.

Expected 5: E1 fails 3 below 5; E2 passes 6 at least 5, suggest E2. Expected 7: both capacities too small, no match. Record actual runs separately and investigate any disagreement.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible **trace**.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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Connect and extend learning

> Connect Build a Paper Prototype to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Build the complete paper sorter: **input** cards, **process** guide, **output** cards, not-sure card, and test log.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 33: Test Normal Cases. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide **trace** column headings and let pupils move a marker while speaking each check aloud. Use not checked for candidates never examined.

Extension: swap partners without author explanation and record whether any **instruction** needs clarification; do not turn one successful handover into a reliability claim.

Assess W4 for a runnable validated **process**, first-match stopping, advancement, exhaustion and separate expected and observed records. The 5-person run should reach E2 and the 7-person run should exhaust both rows.

After support, retry with group size 3, then blank: 3 selects E1 at equality; blank requests clarification before any capacity checks. Ask the pupil to explain the **evidence** rather than only name the result.

Week 33 organises routine cases into a test record with expected and **observed result**s, keeping the **prototype** version fixed during each test set.

Plan the delayed return

Suggested return points in this level: ai-1/33, ai-1/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s.

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Prepare, teach and assess

> Use this preparation summary alongside the detailed Build a Paper Prototype lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Build a runnable paper **prototype** with explicit **instruction**s and visible state

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Use the manual checklist to predict P1's **output** for 11 minutes No. Then ask a partner to operate your saved cards.

Record the **expected result** separately from the observed steps and **output**, including the reason. Explain why the **operator** should stop without checking later rows.

Supported practice answer

Reasoned practice answer: The expected **output** is B1: 10 is no more than 11, quiet is unrestricted and B1 is first. A correct observed **trace** validates, starts at B1, passes both checks and suggests B1 with this reason.

Stop because the specification selects the first fit. If a pupil observes a different run, retain and investigate that **evidence** rather than replacing it with this expected **trace**.

Check every learner before discussion: Use the worked case for Building and tracing a paper **prototype**. Proposed explanation: "A correct-looking answer is enough". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Inspect the request, versions and **trace** to check how it was reached.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Insert “if capacity is below count, continue” before accepting a bin. Require a **prediction** and a reason before comparing. **Expected result**: The revised **trace** rejects X and accepts Y for 4.

Reason: 3<4 is true; 7<4 is false, so Y fits.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

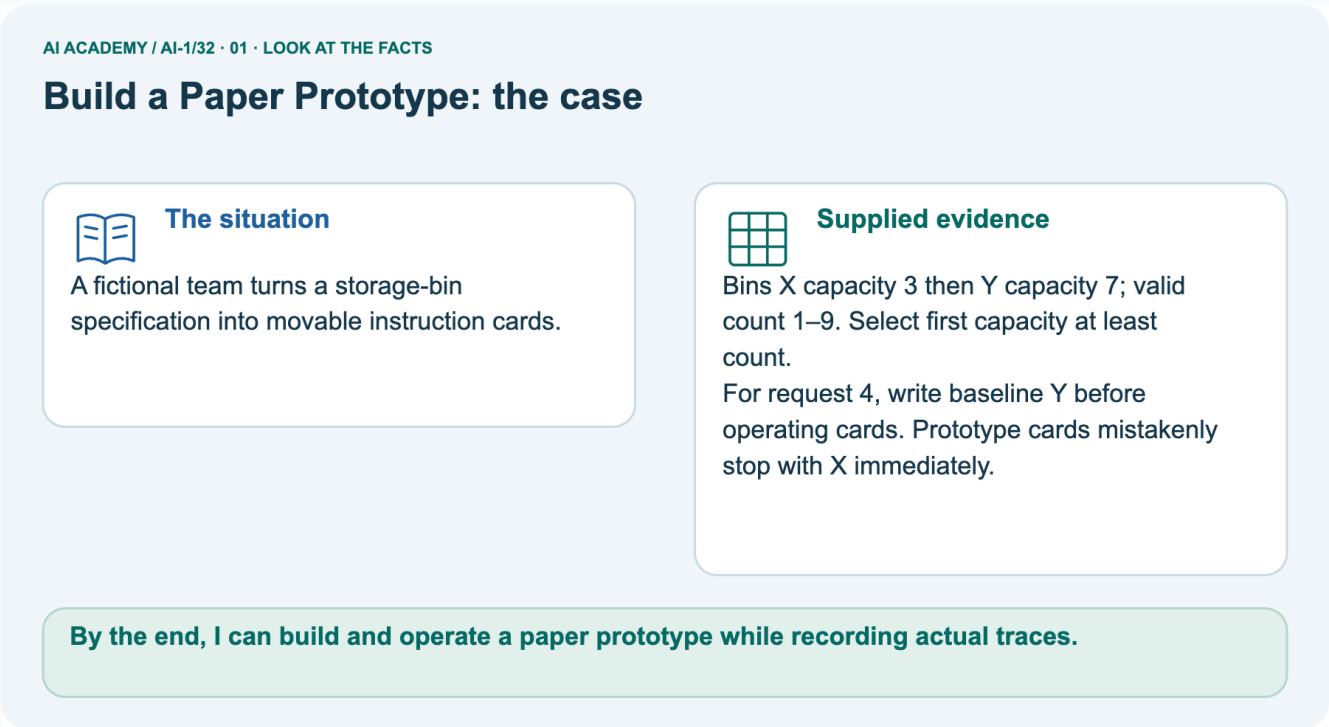
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, **trace**/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Build a Paper Prototype graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Build a Paper Prototype: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional team turns a storage-bin specification into movable **instruction** cards. Supplied facts: Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.

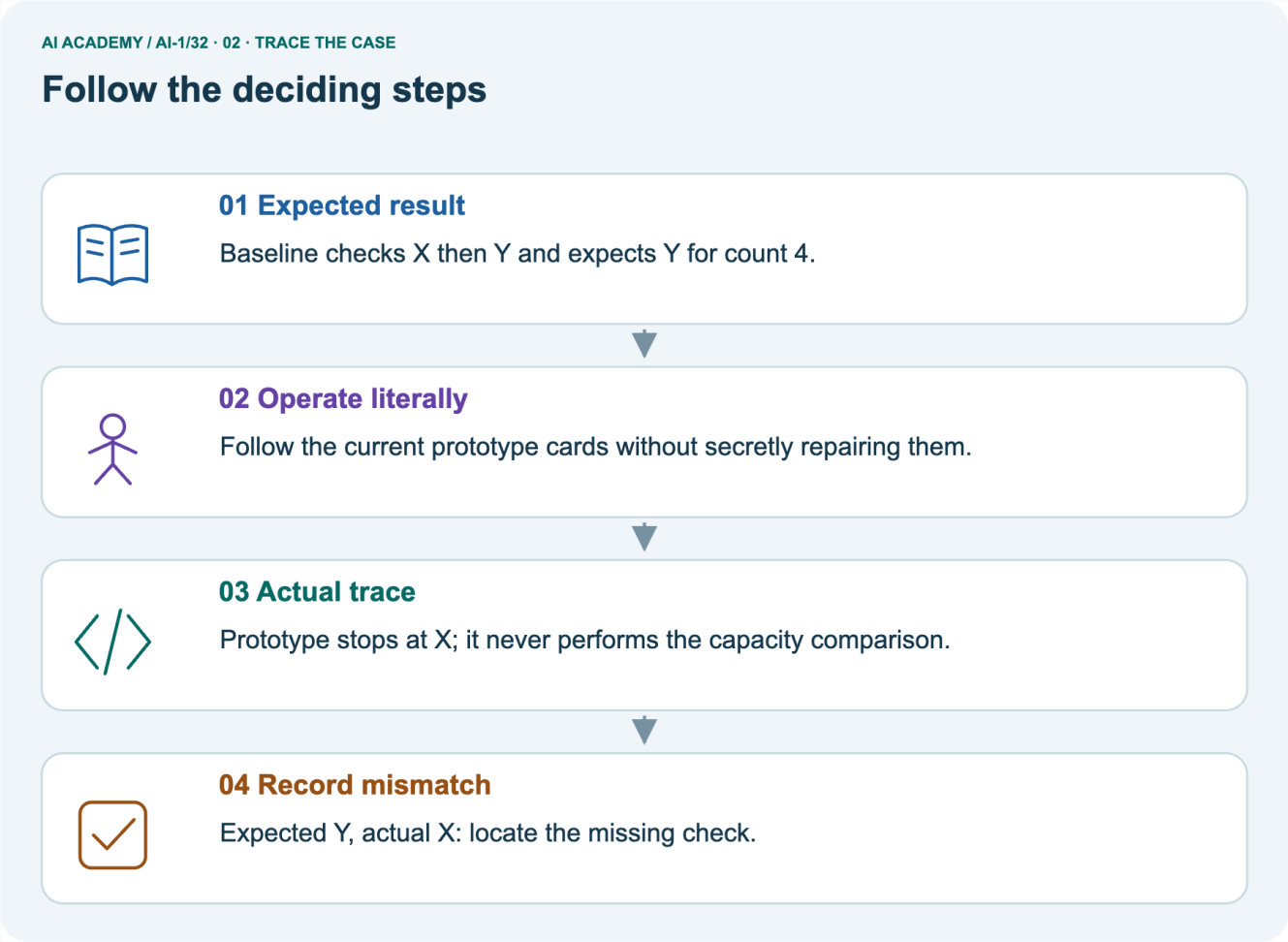
For request 4, write baseline Y before operating cards. **Prototype** cards mistakenly stop with X immediately.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



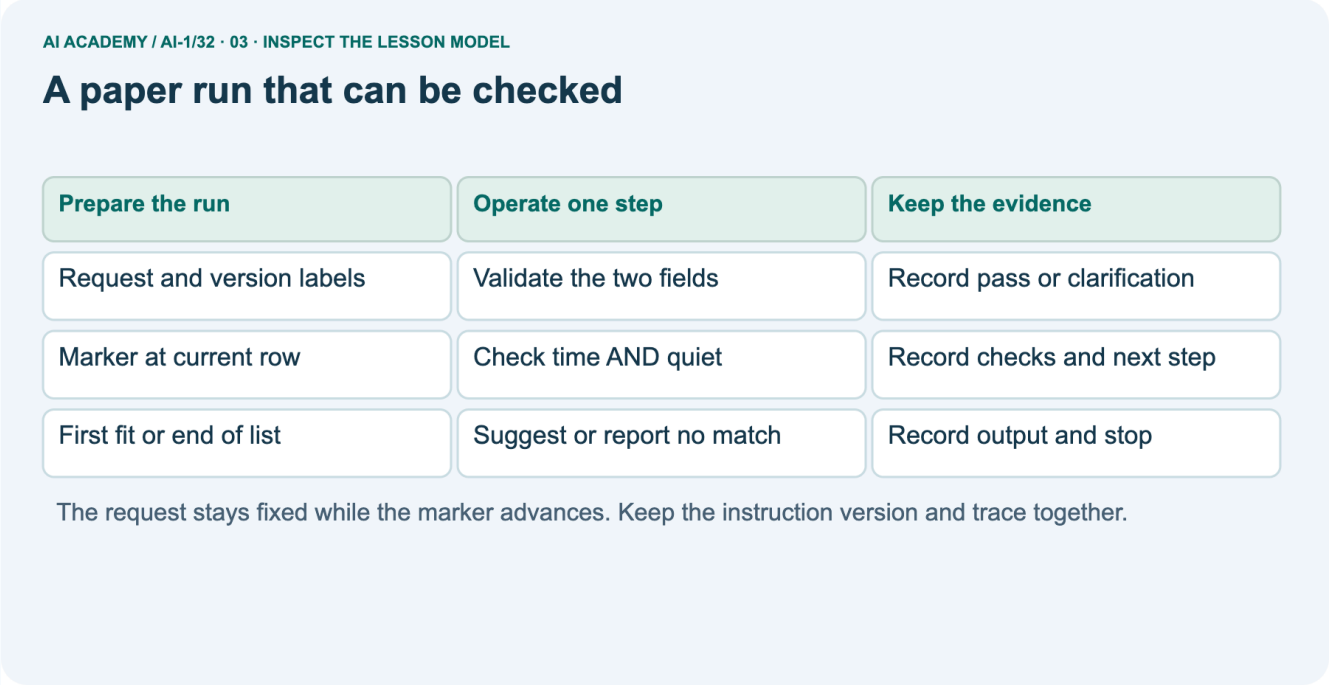
Follow the labelled picture with the matching explanation. Open the vector diagram

Expected result: Baseline checks X then Y and expects Y for count 4. → Operate literally: Follow the current **prototype** cards without secretly repairing them. → Actual **trace**: **Prototype** stops at X; it never performs the capacity comparison. → Record mismatch: Expected Y, actual X: locate the missing check.

> Point to each numbered stage and explain which recorded **input** it uses.

> ### Discuss while showing the picture
>
> Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. A paper run that can be checked



Follow the labelled picture with the matching explanation. Open the vector diagram

A paper run that can be checked. Request and version labels | Validate the two fields | Record pass or clarification. Marker at current row | Check time AND quiet | Record checks and next step.

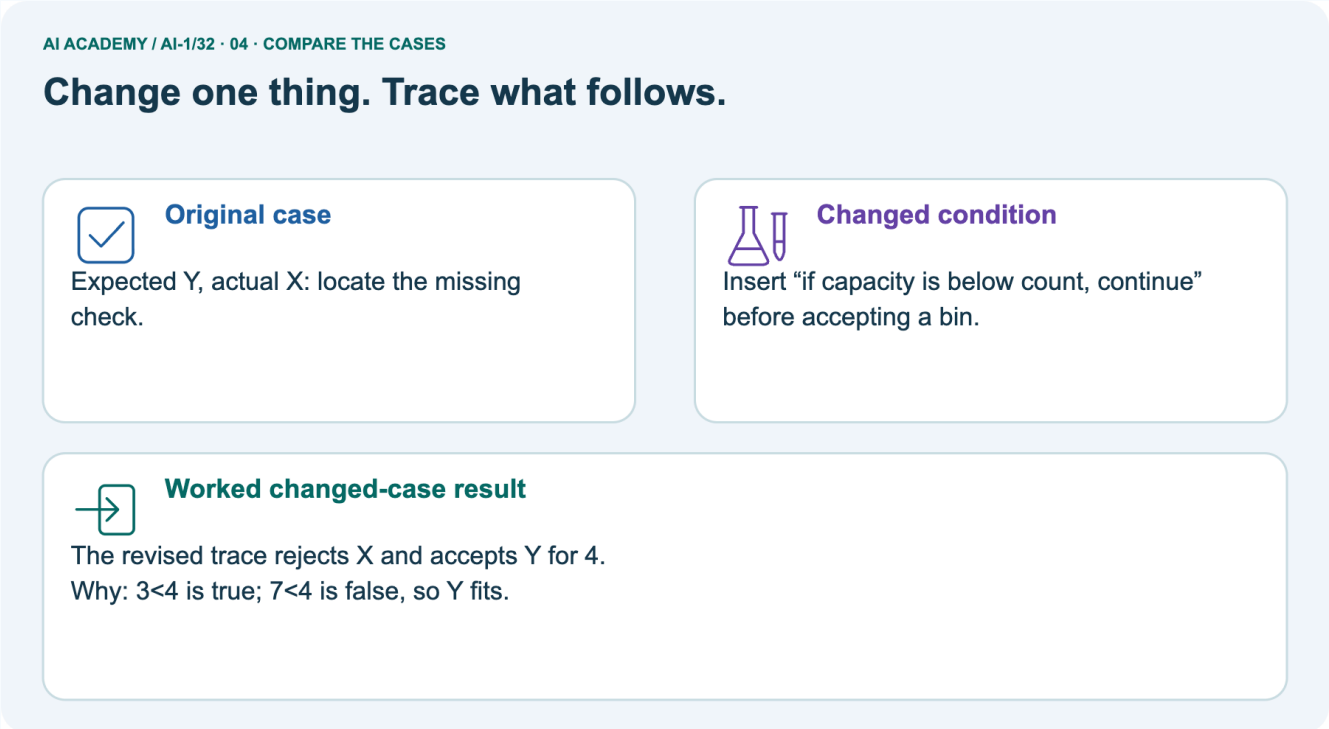
First fit or end of list | Suggest or report no match | Record **output** and stop

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Expected Y, actual X: locate the missing check. Changed condition: Insert “if capacity is below count, continue” before accepting a bin. **Expected result**: The revised **trace** rejects X and accepts Y for 4.

Reason: 3<4 is true; 7<4 is false, so Y fits.

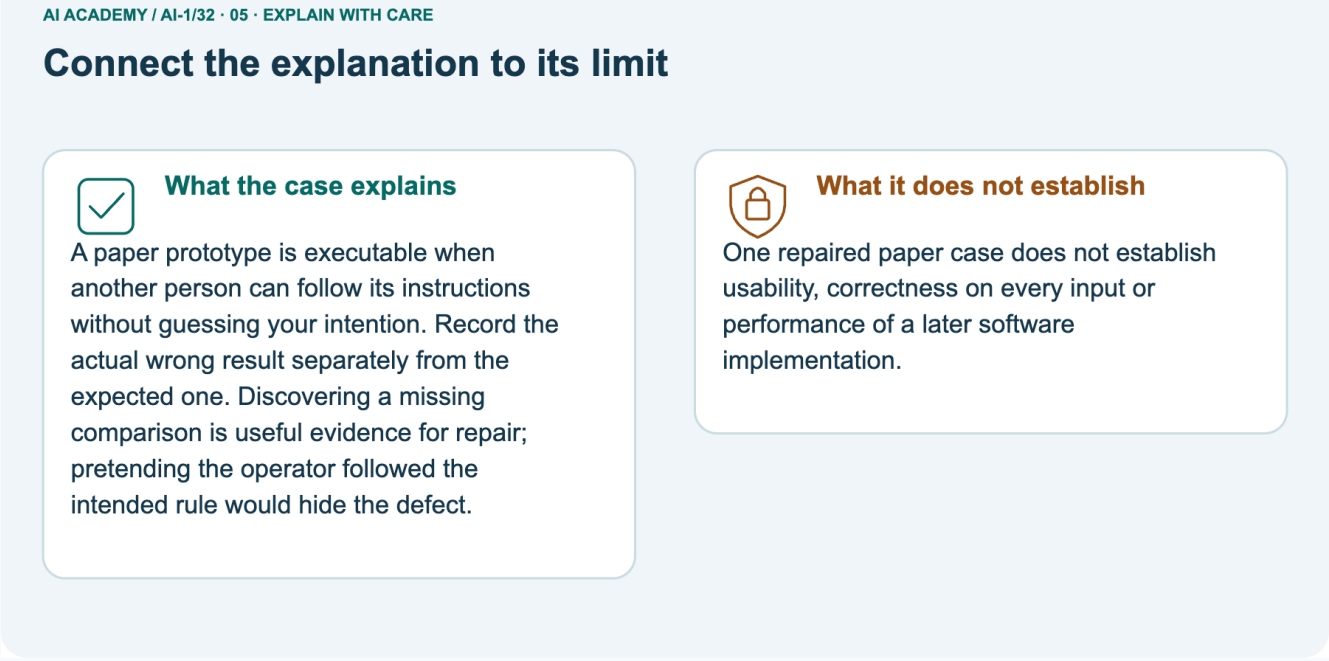
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 3<4 is true; 7<4 is false, so Y fits.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

Limit: One repaired paper case does not establish usability, correctness on every **input** or performance of a later software implementation.

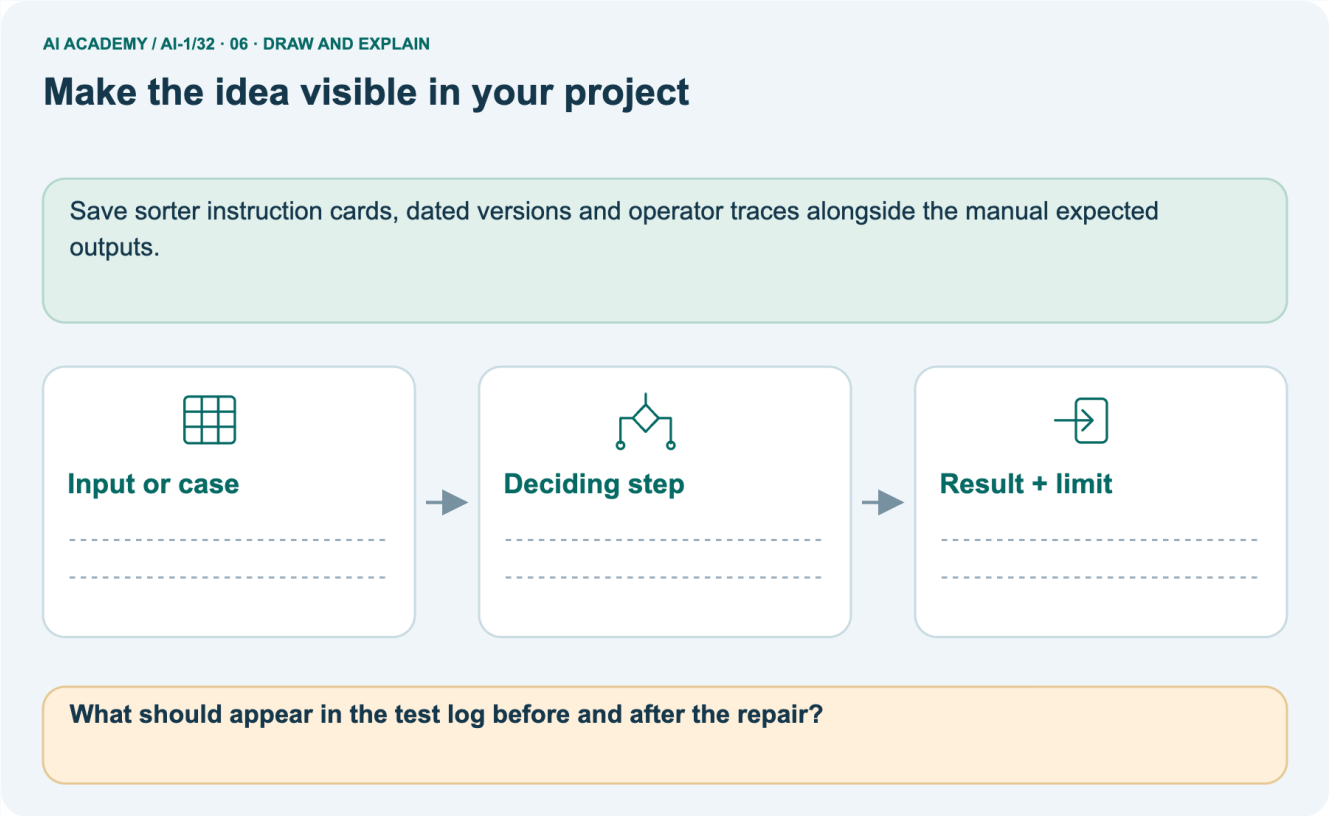
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: One repaired paper case does not establish usability, correctness on every **input** or performance of a later software implementation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s. Draw your **input**, deciding step, result and limit.

Supported question: What should appear in the test log before and after the repair?

Draw your project **trace** and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project **trace** and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Keep original version: expected Y, actual X, missing comparison. Add revised version: expected Y, actual Y, with its full **trace**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.